



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-031 Practice Pack

14 independent attempts | append-only evidence | answer-free task contracts

Artifact	C3_031_TRANSFORMATION_ENGINEERING_WAREHOUSING_PRACTICE_PACK_RC1
Status	2026 - LEARNER EDITION
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation
-> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Practice rules

Use the learner book for the guided walkthrough, then close/minimize it. Pxx is the first genuine attempt. Save what you tried before opening review. Each task must include result + independent validation + one failure/recovery note where applicable.

The starter project is intentionally incomplete. Do not treat TODO scaffolds as hidden solutions.





P01 - ETL vs ELT and transformation-layer boundaries

FIRST ATTEMPT - MATERIAL CLOSED

Draw a fresh source->raw->staging->business transform->mart boundary for returns data; defend replay point.

SAVE BEFORE REVIEW

Create `evidence/L01/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Boundary, grains and replay rule testable.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R01.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P02 - Production SQL style: readability and determinism

FIRST ATTEMPT - MATERIAL CLOSED

Implement deterministic latest-row logic for a changed tied customer-contact fixture.

SAVE BEFORE REVIEW

Create ``evidence/L02/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

One winner/key; tie-break control passes.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R02.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.



P03 - dbt project anatomy and dependency graph

FIRST ATTEMPT - MATERIAL CLOSED

From clean starter, run `version/debug/parse/ls` and add a tiny `ref()`-based model.

SAVE BEFORE REVIEW

Create ``evidence/L03/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

DAG dependency visible; no hardcoded managed model path.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R03.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.



P04 - Sources, staging models and naming contracts

FIRST ATTEMPT - MATERIAL CLOSED

Replace one staging **SELECT *** with explicit columns/types; preserve delivered-version grain.

SAVE BEFORE REVIEW

Create ``evidence/L04/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Source/staging rows reconcile.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R04.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.



P05 - Model materializations and rebuild strategy

FIRST ATTEMPT - MATERIAL CLOSED

Choose materializations for 3 changed models and perform one view/table experiment.

SAVE BEFORE REVIEW

Create ``evidence/L05/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Behavior observed; rebuild path stated.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R05.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.



P06 - Data tests: uniqueness, not-null, relationships and custom assertions

FIRST ATTEMPT - MATERIAL CLOSED

Add grain/not-null/relationship/custom tests; inject one defect, prove fail, repair.

SAVE BEFORE REVIEW

Create `evidence/L06/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Test detects intended defect and returns green after repair.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R06.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P07 - Incremental models and merge/upsert reasoning

FIRST ATTEMPT - MATERIAL CLOSED

Implement base->delta->same-delta rerun logic for orders.

SAVE BEFORE REVIEW

Create `evidence/L07/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Update+insert once; unchanged rerun stable.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R07.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P08 - Snapshots and SCD change-history patterns

FIRST ATTEMPT - MATERIAL CLOSED

Create customer history from v1->v2 and prove current + historical truth.

SAVE BEFORE REVIEW

Create `evidence/L08/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Only legitimate versions exist.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R08.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P09 - Macros, templating and reusable transformation logic

FIRST ATTEMPT - MATERIAL CLOSED

Build one macro, compile it and compare compiled SQL to expected expression.

SAVE BEFORE REVIEW

Create `evidence/L09/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Known-row and edge-case controls pass.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R09.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P10 - Documentation, model descriptions and lineage artifacts

FIRST ATTEMPT - MATERIAL CLOSED

Document one mart with grain/keys/purpose/tests/caveat and trace lineage.

SAVE BEFORE REVIEW

Create `evidence/L10/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Another engineer can understand/rebuild.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R10.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P11 - Dimensional model implementation and grain enforcement

FIRST ATTEMPT - MATERIAL CLOSED

Implement/reconcile order-line fact and two dimensions; inject duplicate dimension key.

SAVE BEFORE REVIEW

Create `evidence/L11/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Fan-out detected before DISTINCT; repaired at source/dimension contract.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R11.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P12 - Surrogate keys, late dimensions and referential integrity

FIRST ATTEMPT - MATERIAL CLOSED

Process late C999 fact through Unknown->resolved lifecycle.

SAVE BEFORE REVIEW

Create `evidence/L12/attempt\_001/`. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

No fact loss; unresolved monitor 1->0.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R12.

AFTER REVIEW

Close review and do its changed retry in `retry\_001/`. Mastery depends on that changed result, not recognition of the review answer.



P13 - Performance-aware SQL and query-plan reasoning

FIRST ATTEMPT - MATERIAL CLOSED

EXPLAIN a non-trivial query, make one safe change, prove equivalence.

SAVE BEFORE REVIEW

Create ``evidence/L13/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Plan observation saved; outputs equivalent.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R13.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.



P14 - CI checks and curated serving-model handoff

FIRST ATTEMPT - MATERIAL CLOSED

Write CI-safe check sequence + serving-model handoff for a changed mart.

SAVE BEFORE REVIEW

Create ``evidence/L14/attempt_001/``. Save code/config used, actual output/error, one independent validation, confidence 0-4, and your main uncertainty. Do not overwrite this folder.

SUCCESS CHECK

Blocking checks, no secrets, clear rebuild contract.

IF BLOCKED

Use the lesson help/recovery ladder only. Save where you got stuck before opening R14.

AFTER REVIEW

Close review and do its changed retry in ``retry_001/``. Mastery depends on that changed result, not recognition of the review answer.